Draft Scheme of Valuation/Answer Key

(Scheme of evaluation (marks in brackets) and answers of problems/key)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER M.C.A.DEGREE EXAMINATION, DECEMBER 2021

Course Code: 20MCA201

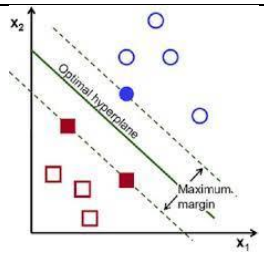
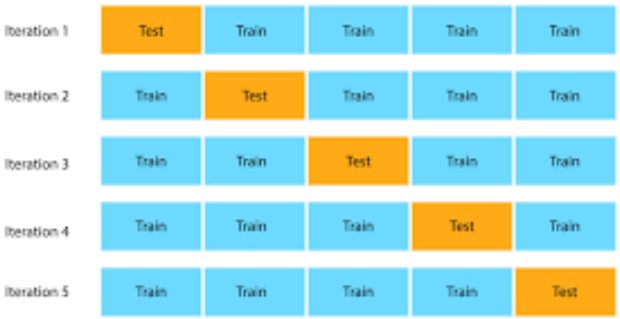
Course Name: DATA SCIENCE AND MACHINE LEARNING

Max. Marks: 60		Duration: 3 Hours

PART A

		Answer all questions, each carries3 marks.	Marks
1		Data science is a subset of AI, and it refers more to the overlapping areas of statistics, scientific methods, and data analysis—all of which are used to extract meaning and insights from data.Data science is a process that empowers better decision-making through interpreting, modeling, and deployment. The reason why we need data science is the ability to process and interpret data. This enables companies to make informed decisions around growth, optimization, and performance. What is data science -1.5 marks Need for data science-1.5 marks	(3)
2		Types of data- -Numeric or Continuous -Categorical or Nominal 1 Mark for each type of data Examples 0.5 Mark each	(3)
3		A supervised learning algorithm learns from labeled training data, helps you to predict outcomes for unforeseen data. Unsupervised learning is a machine learning technique, where you do not need to supervise the model. Instead, you need to allow the model to work on its own to discover information. It mainly deals with the unlabelled data. Explanation & one example for supervised machine learning algorithm-1.5 marks Explanation & one example for unsupervised machine learning algorithm -1.5 marks	(3)

4		Strengths:Simple & effective, Makes no assumptions about the underlying data distribution,Fast training phase Weaknesses : oDoes not produce a model, limiting the ability to understand how the features are related to the class o Requires selection of an appropriate k o Slow classification phase Any 3 strengths -1.5 marks Any 3 weaknesses -1.5 marks	(3)
5		Pruning reduces the size of decision trees by removing parts of the tree that do not provide power to classify instances. -1 mark i)post-pruning -1 mark ii)pre-pruning- 1 mark	(3)
6		Ordinary least squares (OLS) regression is a statistical method of analysis that estimates the relationship between one or more independent variables and a dependent variable; the method estimates the relationship by minimizing the sum of the squares in the difference between the observed and predicted values of the dependent variable configured as a straight line. The Least Squares Regression Line is the line that makes the vertical distance from the data points to the regression line as small as possible. It's called a "least squares" because the best line of fit is one that minimizes the variance (the sum of squares of the errors).	(3)
7		Definition 1 Mark Activation function examples 1 mark each	(3)
8		The distance between the line and the closest data points is referred to as the margin. The best or optimal line that can separate the two classes is the line that has the largest margin. This is called the Maximal-Margin hyperplane.The margin is calculated as the perpendicular distance from the line to only the closest points. Only these points are relevant in defining the line and in the construction of the classifier. These points are called the support vectors.	(3)

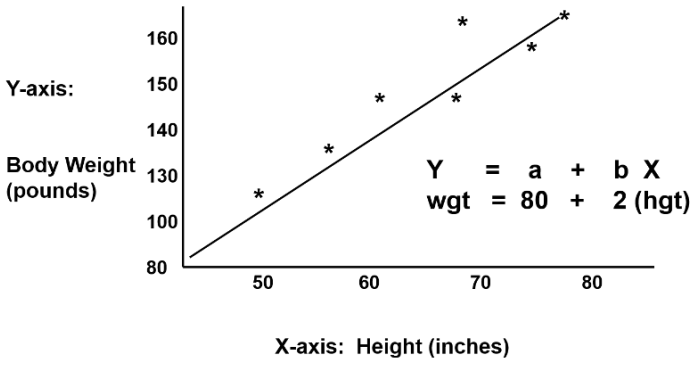
		 Explanation 1.5 marks, Figure 1.5 marks	
9		K-Fold CV is where a given data set is split into a *K* number of sections/folds where each fold is used as a testing set at some point. For example, in a 5 fold CV, the data set is split into 5 folds. In the first iteration, the first fold is used to test the model and the rest are used to train the model. In the second iteration, 2nd fold is used as the testing set while the rest serve as the training set. This process is repeated until each fold of the 5 folds have been used as the testing set. 	(3)
10		A bootstrap sample is a smaller sample that is "bootstrapped" from a larger sample. Bootstrapping is a type of resampling where large numbers of smaller samples of the same size are repeatedly drawn, with replacement, from a single original sample.	(3)
		PART B	
		Answer any one question from each module. Each question carries 6 marks.	
		Module I	
11		Data Preparation - Data Exploration, Data Quality, Missing Values, Data Types and Conversion, Transformation, Feature Selection, Data Sampling. Any 4 processes with explanation-1.5 marks each.	(6)
		OR	
12		Mean & standard deviation formulae & correct answer - 3 Marks each mean 34.53 megapascals, standard deviation 0.07474 megapascals	(6)
		Module II	
13		Step1: Choose K=3 Step2:Calculate distance	(6)

X	Y	Classification	Distance
8	6	Outstanding	3.16
5	6	Good	1
7	3	Good	4.47
6	9	Outstanding	2.23

Step 3:Sort the distances in ascending order

Step4 : Among the 3-nearest neighbours, the class label which occurs

		most frequently is "**Good**". Step 5: Assign the class label **Good** to the test instance. Steps - 4 marks , correct answer-2 marks	
		OR	
14		Features are F1 = "Swim", F2 = "Fly", F3 = "Crawl". The class labels are c1 = "Animal", c2 = " Bird", c3 = "Fish". The test instance is (Slow, Rarely, No) and so we have: x1 = "Slow", x2 = "Rarely", x3 = "No". Step 1:Compute following probabilities. P(c1) = No. of records with class label "Animal" /Total number of examples = 5/12 P(c2) = No. of records with class label "Bird"/ Total number of examples = 4/12 P(c3) = No of records with class label "Fish"/Total number of examples = 3/12 Step 2.The conditional probabilities are calculated as follows: P((Fl = Slow)|cl) = No. of records with Fl = Slow and class label cl/ No. of records with class label cl = 2/5. Step 3:Now calculate the following numbers: q1 = P(xl|cl)P(x2|cl)P(x3|cl)P(cl) = (2/5) × (1/5) × (3/5) × (5/12) = 0.02 q2 = P(xl|c2)P(x2|c2)P(x3|c2)P(c2) = (0/4) × (0/4) × (3/4) × (4/12) = 0 q3 = P(xl|c3)P(x2|c3)P(x3|c3)P(c3) = (2/3) × (0/3) × (3/3) × (3/12) = 0 Step 4:Now max{q1, q2, q3} = 0.05. Step 5:The maximum is q1 and it corresponds to the class label c1 = " Animal". So we assign the class label "Animal" to the test instance "(Slow, Rarely, No)". For steps -4 marks , correct answer-2 marks	(6)
		Module III	

15	a)	There are three positive examples and three negative examples. Thus,P (+) =3/6 and P(−) = 3/6. The entropy of the classification is −3/6 log2(3/6) − 3/6 log2(3/6) = 1 (3 Marks)	(6)
	b)	The entropy for a2 is given by −4/6 log2(4/6) − 2/6 log2(2/6) =0.9186 Therefore, the information gain for a2 is 1- 0.9186 = 0.0814. (3 Marks)	
		OR	
16		Regression analysis makes use of mathematical models to describe relationships. Explanation for Simple linear regression model, multiple linear regression model, regression coefficients,ordinary least squares estimation. Regression Formula: $Y = a + bX$ where slope of trend line is calculated as: $b_1 = \frac{\sum (x - \bar{x}) * (y - \bar{y})}{\sum (x - \bar{x})^2}$ and the intercept is computed as: $b_0 = y - (b_1 * X)$ (4 Marks) Suppose that height was the only determinant of body weight. If we were to plot height (the independent or 'predictor' variable) as a function of body weight (the dependent or 'outcome' variable), we might see a very linear relationship, as illustrated below.  We could also describe this relationship with the equation for a line, Y = a + b(x), where 'a' is the Y-intercept and 'b' is the slope of the line. We could use the equation	(6)

		to predict weight if we knew an individual's height. In this example, if an individual was 70 inches tall, we would predict his weight to be: Weight = 80 + 2 x (70) = 220 lbs. Example-2 Marks	
		Module IV	
17		Back-propagation, short for "backward propagation of errors," is **an algorithm for supervised learning of artificial neural networks using gradient descent**. Explanation with the help of a figure 3 marks. Backpropagation algorithm -3 Marks	(6)
		OR	
18	a) b)	Explanation-2 marks , example - 1 mark Kernel function-1 Mark. Explanation of the kernel trick to construct a classifier for a dataset that is not linearly separable- 2 Marks	(6)
		Module V	
19		Labelling TP,FN,FP,TN values in confusion matrix -2 Marks. Precision & Recall calculation 2 marks each. TP = 40, FN = 15,FP = 10,TN = 25 Precision=TP/(TP +FP) =40/50 = 0.8 Recall=TP/(TP+FN) =40/55 = 0.727	(6)
		OR	
20		Bagging 3 Marks Boosting 3 Marks.	(6)
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